

Corroborate: Independence-Aware Event Corroboration Scoring

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Abstract

Open-source event monitoring drowns in *reports* but is starved of *confidence*. A magnitude-6 earthquake, a strike, or a wildfire generates a flood of claims — seismic bulletins, news wires, social posts — and the operative question is not “did anyone say it happened?” but “**how much should I believe it happened, given who is saying so?**” Existing tooling answers this manually (human verifiers) or not at all (raw aggregation).

My solution is **Corroborate**, a pipeline that answers this automatically and *quantitatively*: it associates heterogeneous claims in space and time, discounts non-independent sources, and emits a **calibrated probability** that each candidate event is real, together with a **refutation flag** when the claims contradict each other. To test the veracity of the pipeline I calibrated it using earthquakes, for which USGS provides free, authoritative ground truth. From here I can derive a score that represents a clear confidence signal and indicates strong cross-modal corroboration (social media independently confirming a seismic network).

1. The problem

Three observations motivated me here:

1. **Count is not confidence.** Twelve accounts reposting one news wire is one piece of evidence, not twelve. Any system that scores confidence by counting reports is fooled by amplification.
2. **Verification should be continuous, not binary.** A “verified / unverified” stamp throws away information. A *calibrated* probability — where a score of 0.8 means roughly 80% of such events turn out real — is both more honest and more useful downstream.
3. **Disagreement is signal.** When sources place “the same” event in two incompatible locations, averaging them produces a confident-looking centroid that is simply wrong. Contradiction should be surfaced, not smoothed away.

Corroborate is built around these three principles: it should be **independence-aware**, **calibrated**, and **refutation-aware**.

1.1 Relation to prior work

Operational OSINT tools split into two families. **Human-verification** projects (GeoConfirmed, Bellingcat) geolocate and confirm content by hand — accurate but unscalable and binary. **Aggregation** feeds (Liveuamap, ACLED) collect and display many reports, with cross-referencing done by editors, not algorithms.

The closest work I have found discussing *automated* event corroboration is **EventMapper** (arXiv:2001.08700), which is a proposed framework that scores corroboration on a spatio-temporal grid. I build upon this with three components it lacks: explicit **source-independence weighting**, **probability calibration** against ground truth, and a **refutation** signal.

2. Data model

Every input, regardless of source, normalizes to a **Claim**: an assertion that *something happened at a place and time*, with explicit uncertainty.

field	meaning
source_id, source_type	origin (usgs, emsc, mastodon, ...; seismic_network / social / news)
event_time, time_uncertainty_s	claimed time and its uncertainty
lat, lon, loc_uncertainty_km	claimed location and its uncertainty
magnitude	optional intensity
raw_text, external_id	text + dedup key

A **candidate Event** is a cluster of Claims, carrying aggregate quantities (n_claims , $n_independent$, n_source_types), a calibrated score, and a refutation_flag. Sources designated as *ground truth* (here, USGS) are held out of the candidate-forming pool entirely and used only as calibration labels — this prevents the label from leaking into the features.

3. The pipeline

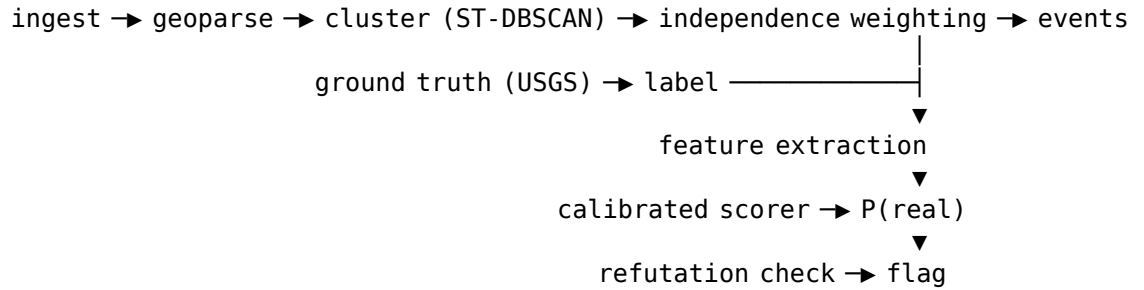


Figure 1. End-to-end dataflow. The authoritative ground-truth source (USGS) is held out as calibration labels and never enters the candidate pool; the probabilistic pool — other networks plus social — is what gets clustered and scored. This separation is what keeps the label from leaking into the features.

3.1 Ingest

One poller per source emits normalized Claims. Authoritative networks (USGS, EMSC) arrive geocoded; social/news sources do not. Pollers are fault-isolated — one feed failing or lacking credentials never aborts the run — and dedup-on-write by ($source_id$, $external_id$) makes repeated polling idempotent.

3.2 Geoparse

Social and news claims carry only free text. I resolve a location with an **offline gazetteer** ($\approx 25k$ cities and country capitals) augmented by a small supplement of sub-national seismic regions (US states, Honshū, Sumatra, ...). Two guards keep precision high:

- a **proper-noun heuristic** — only capitalized spans are matched, so the common word “*much*” never resolves to the town of Much, Germany;
- a **confidence floor** — if no place matches, the claim is **dropped**, not guessed. A missing coordinate is recoverable; a wrong one silently corrupts every downstream stage.

Resolved locations carry an uncertainty radius (a named town ≈ 25 km, a country ≈ 400 km) used later as a soft weight.

3.3 Independence weighting — the core idea

This is the stage that separates corroboration from amplification. Within a set of co-located, contemporaneous claims I detect **near-duplicate text** and collapse it, so reposts of a single wire count once.

Each claim’s text is shingled into overlapping 3-word tokens and fingerprinted with **MinHash** (128 permutations). Pairs whose estimated Jaccard similarity exceeds a threshold $\tau = 0.80$ are linked, and a union-find pass groups the transitive closure into near-duplicate groups $G_1 \dots G_k$.

Each group contributes a total of one unit of evidence, split evenly across its members:

$$w_i = \frac{1}{|G(i)|}, \quad n_{\text{independent}} = k$$

where w_i is the independence weight of claim i , $|G(i)|$ the size of the near-duplicate group containing it, k the number of such groups in the cluster, and $n_{\text{independent}}$ the effective independent-source count.

So twelve identical reposts form one group of twelve: each member gets a weight of $1/12$ and the cluster’s independent-source count is 1, not 12. Twelve distinct eyewitness texts form twelve groups and count as twelve. Claims with too little text to shingle each form their own singleton group rather than colliding on an empty fingerprint. Crucially, this runs *within a spatio-temporal cluster* — the clusters formed in §3.4 — so “near-duplicate text” reliably means “the same wire, reposted” rather than a coincidence across unrelated events.

Formally, each claim’s text is reduced to a set of word-shingles, and the **Jaccard similarity** of two shingle sets A, B is

$$J(A, B) = \frac{|A \cap B|}{|A \cup B|}.$$

where A and B are the word-shingle sets of two claims, $|A \cap B|$ the shingles they share, and $|A \cup B|$ their total distinct shingles; $J \in [0, 1]$, with 1 meaning identical text.

MinHash estimates J from compact signatures, avoiding the $O(n^2)$ exact set comparisons. Two claims are linked when their estimated similarity $\geq \tau$, and union-find takes the transitive closure into groups $G_1 \dots G_k$. The per-claim weights then satisfy

$$\sum_i w_i = \sum_{g=1}^k \sum_{i \in G_g} \frac{1}{|G_g|} = k = n_{\text{independent}},$$

where G_g is the g -th near-duplicate group and $|G_g|$ its size; each group contributes exactly 1, so the k groups sum to k .

so “total evidence mass” and “independent-source count” coincide by construction.

Worked example. Suppose six claims survive into one Mindanao cluster:

claim	source	text	group	weight
c1	EMSC	“M6.8 Mindanao, Philippines”	G ₁	1
c2	Mastodon	“earthquake in Mindanao, felt it just now”	G ₂	1
c3	Mastodon	“BREAKING: 6.8 quake strikes Mindanao”	G ₃	1/3
c4	Mastodon	“BREAKING: 6.8 quake strikes Mindanao”	G ₃	1/3
c5	Bluesky	“BREAKING: 6.8 quake strikes Mindanao”	G ₃	1/3
c6	Mastodon	“mindanao earthquake”	G ₄	1

c3-c5 are reposts of one wire (Jaccard $\approx 1 \geq \tau$) and collapse into a single group G₃; the rest are textually distinct (Jaccard $< \tau$) and stand alone. A naïve report count claims **6** corroborating sources. Independence weighting instead gives

$$n_{\text{independent}} = |\{G_1, G_2, G_3, G_4\}| = 4,$$

because the three-fold amplified wire contributes one unit of evidence ($3 \times \frac{1}{3} = 1$), not three. The same weights drive the cluster centroid, so the reposts cannot pull the estimated location toward whichever wire was loudest. That gap — **6 raw vs 4 effective** — is precisely the amplification the score must refuse to reward.

3.4 Spatio-temporal clustering

Candidate events are formed by **ST-DBSCAN**. I build a pairwise distance matrix of great-circle (haversine) distances and **mask** any pair whose time gap exceeds the temporal window to infinity (a large sentinel in practice):

$$D_{ij} = \begin{cases} \text{haversine}(c_i, c_j) & |t_i - t_j| \leq \varepsilon_t \\ \infty & \text{otherwise} \end{cases}$$

where D_{ij} is the distance DBSCAN sees between claims i and j , c_i is claim i 's location and t_i its claimed time, ε_t the temporal window (30 min), $\text{haversine}(\cdot)$ the great-circle distance in km, and ∞ marks a pair too far apart in time to ever be neighbours.

DBSCAN then runs on this precomputed matrix with a spatial radius $\varepsilon_s = 100 \text{ km}$, temporal window $\varepsilon_t = 30 \text{ min}$, and $\text{min_samples} = 2$. The mask makes a pair count as neighbours only if they are close in **both** space and time. Points DBSCAN would discard as noise are *kept* as singleton candidates — an uncorroborated report is still a candidate event; it simply scores low. Each cluster's centroid and time are independence-weighted means, so a swarm of reposts cannot drag the location toward whichever source was loudest.

Formally, a claim p is a **core point** when at least min_samples claims fall within ε_s of it under the masked metric D ; clusters are the maximal density-connected sets of core points and their neighbours, and everything else becomes a singleton candidate. Each cluster's reported location and time are the independence-weighted means

$$\bar{c} = \frac{\sum_i w_i c_i}{\sum_i w_i}, \quad \bar{t} = \frac{\sum_i w_i t_i}{\sum_i w_i},$$

where $\bar{c}$ and $\bar{t}$ are the cluster's reported location and time, c_i and t_i are claim i 's location and time, w_i its independence weight from §3.3, and the sums run over the cluster's member claims.

reusing the weights from §3.3 so amplified reposts neither shift the centroid nor inflate the apparent spatial agreement.

3.5 Features and scoring

Each candidate event is summarized by seven features:

feature	intuition
n_independent	effective independent sources (post-dedup)
n_source_types	cross-modal diversity (seismic + social + news)
spatial_dispersion_km	how tightly the claims agree on location
temporal_spread_s	how tightly they agree on time
consolidation_lag_s	how quickly corroboration accrued
max_source_prior	reliability of the best contributing source
max_magnitude	strongest reported intensity

The scorer is a `StandardScaler` → `LogisticRegression` pipeline wrapped in scikit-learn’s `CalibratedClassifierCV` (Platt scaling), so its output is an honest probability rather than an uncalibrated margin. Platt scaling is chosen over isotonic because it is robust on the modest label volumes available.

Concretely, on the standardized feature vector $\tilde{x}$ the logistic stage produces a margin

$$z = \beta_0 + \sum_{j=1}^7 \beta_j \tilde{x}_j,$$

where z is the logistic margin (log-odds), $\tilde{x}_j$ the j -th of the seven standardized features, β_j its learned coefficient, and β_0 the intercept.

and Platt scaling maps that margin to a probability through a sigmoid fitted on held-out folds,

$$P(\text{real}) = \sigma(az + b) = \frac{1}{1 + e^{-(az+b)}}.$$

where σ is the logistic sigmoid, z the margin above, and a, b the two Platt-scaling parameters fitted on held-out folds; $P(\text{real}) \in (0, 1)$.

The two-stage fit — logistic, then sigmoid recalibration — is what lets the output be read as a *frequency* (how often such events are real) rather than a mere ranking score.

3.6 Calibration against ground truth

A candidate is labelled **positive** if a held-out USGS ground-truth event lies within $\Delta d \leq 150 \text{ km}$ and $\Delta t \leq 60 \text{ min}$. Because USGS claims never enter the candidate pool, the label is independent of the features. I split candidates by time (70% train / 30% test — a forecasting-style split, not a random one) and report:

- the **reliability diagram** — the headline plot: across score bins, does the predicted probability match the observed fraction of real events?
- the **Brier score** (mean squared error of the probabilities), and
- **PR-AUC** (discrimination under class imbalance).

With predictions $\hat{p}_i$ and binary labels y_i , the Brier score is the mean squared error of the probabilities,

$$\text{BS} = \frac{1}{N} \sum_{i=1}^N (\hat{p}_i - y_i)^2,$$

where N is the number of evaluated events, $\hat{p}_i$ the predicted $P(\text{real})$ for event i , and $y_i \in \{0, 1\}$ its label (1 if a held-out USGS event confirms it); lower is better and 0 is perfect.

and the reliability diagram partitions predictions into bins and plots, for each bin, the mean predicted probability against the observed positive rate. Perfect calibration lies on the diagonal — *mean predicted = fraction real* — so vertical distance from the diagonal is exactly the miscalibration in that score band.

3.7 Refutation

After scoring, each multi-claim event is tested for internal contradiction. I find the farthest-apart pair of member claims; if their separation exceeds $\text{REFUTE_DIST_KM} = 300 \text{ km}$, I assign every claim to its nearer extreme and flag the event when **both** ends carry at least $\text{REFUTE_MIN_MASS} = 30\%$ of the independence-weighted mass:

$$\text{refuted} \iff \max_{i,j} d(c_i, c_j) > 300 \text{ km} \wedge \frac{\min(m_A, m_B)}{m_A + m_B} \geq 0.30$$

where $d(c_i, c_j)$ is the haversine distance between member claims i and j , and m_A, m_B are the independence-weighted masses of the

claims nearest each of the two farthest-apart extremes; the test applies only to clusters of three or more claims.

The mass condition is what distinguishes a genuine two-location contradiction from a single stray outlier: one mislocated claim attached to an otherwise tight cluster carries little mass and does not trip the flag. Refuted events are surfaced as disputed rather than collapsed to a meaningless averaged centroid.

3.8 Continuous operation

The pipeline is built to run unattended on a live feed. Because ST-DBSCAN is $O(n^2)$ in the number of claims, clustering each cycle over the entire accumulated history would not scale; instead every run clusters only a rolling window of recent claims — anchored on the most recent claim, so the window behaves identically on a live feed and on a replayed archive — rebuilds the events inside that window, leaves older events intact, and prunes claims and events past a longer retention horizon. Scoring is decoupled from training: the calibrated model is refit only periodically, while a lightweight pass re-scores the current events from the saved model every cycle. A single lock-guarded step sequences ingest $\rightarrow$ cluster $\rightarrow$ score (the store is single-writer, so runs must not overlap), scheduled to repeat every few minutes with a daily recalibration.

4. Results

Calibrating on live data — USGS `all_day` as ground truth, EMSC as an independent network, and Mastodon as an unauthenticated social feed (Bluesky and X are implemented but credential-gated, and contributed nothing to this run):

- **Ground-truth confirmation (cross-network).** Because USGS is held out as labels, EMSC $\leftrightarrow$ USGS agreement appears not as a shared cluster but as a *label match*: ≈ 59 of ~ 900 candidate events fall within 150 km / 60 min of a held-out USGS event. These positives are what the scorer is calibrated against.
- **Cross-modal corroboration** emerges once the social layer is added: from ~ 125 Mastodon claims, the pipeline surfaced **7 events where social posts and the EMSC seismic network independently confirm the same quake** (e.g. an M3.9 off Central America, an M4.5 near Oaxaca, an M4.1 in Chile).
- The score learns a **sensible, monotonic signal**: estimated $P(\text{real})$ rises with magnitude ($\approx 3\% \rightarrow 18\%$ across magnitude

bands), and the top reliability bin is $\approx 60\%$ observed-real — i.e. higher scores really do correspond to more-real events.

- **Refutation** stays correctly silent on this data: the largest live cluster spans 169 km, below the 300 km threshold, so nothing is flagged. The detector is validated on a synthetic 480 km chain that it correctly flags.
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5. Limitations

I state these plainly because they bound the current claims.

1. **Absolute calibration needs noisy sources.** With only authoritative seismic networks, almost every candidate is real, so there are few negatives to calibrate against. The social layer supplies genuine false candidates (geoparse coincidences, posts about old quakes) and is what makes calibration meaningful; calibration sharpens as social volume grows.
 2. **Temporal base-rate shift.** USGS’s 24-hour window overlaps only part of EMSC’s span, concentrating positives in the most recent events; under a time-ordered split this makes the held-out test *under-confident* in absolute terms even when ranking is good. Matching the feed windows resolves it.
 3. **Geoparse recall.** A gazetteer with a proper-noun guard favours precision over recall — it will miss colloquial or sub-national place references that a learned NER model would catch.
 4. **Swarm over-merging.** A dense aftershock sequence can merge into a single over-large cluster; magnitude- and text-aware splitting is future work.
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6. Generalization and future work

Earthquakes are a *calibration substrate*, not the goal: they are the rare domain with free, exact, independent ground truth, which makes them ideal for proving the scorer is honest before pointing it at domains where truth is scarce — conflict events, disasters, infrastructure outages. The independence weighting, clustering, calibration, and refutation are all domain-agnostic; only the ingest adapters and the gazetteer change.

Planned directions: **cross-modal corroboration as a first-class feature** (does a social claim line up with a satellite or ADS-B signal?); a **temporal analogue of refutation** (incompatible times,

not just locations); **learned source priors** from historical accuracy rather than hand-set values; and **penalizing**, not merely flagging, the scores of refuted events.

7. Conclusion

Corroborate reframes event verification as a calibrated estimation problem. By counting *independent* corroboration rather than raw reports, calibrating the resulting score against ground truth, and explicitly flagging contradiction, it turns a flood of heterogeneous claims into a single, honest number — and does so automatically. Validated on live earthquake data, it recovers a monotonic, well-ranked confidence signal and surfaces real cross-modal corroboration, while remaining transparent about exactly where its current confidence is and is not trustworthy.